République Tunisienne
Ministère de l'Enseignement Supérieur et de la Recherche Scientifique
Direction Générale des Études Technologiques

Institut Supérieur des Études Technologiques de Nabeul

COMPTE RENDU DE TRAVAUX PRATIQUES

Module : Atelier Technologies WAN

TP 4 : Tunnel GRE over IPsec

Tunnel GRE entre deux sites puis chiffrement par IPsec – réalisation sous GNS3

Réalisé par	Amin MRIRI
Classe	RSI31
Enseignant	M. Zied Romdane
Date du TP	11/11/2026
Outils	GNS3 2.2.61 – Cisco 3725

Année universitaire 2026 – 2027

1. Objectifs

Relier deux sites (10.0.10.0/24 et 10.0.20.0/24) à travers un réseau public (routeur WAN) avec un **tunnel GRE**, puis **chiffrer ce tunnel avec IPsec**. GRE transporte tout type de trafic, y compris le multicast des protocoles de routage, mais ne chiffre rien ; IPsec chiffre mais ne transporte que de l'unicast IP. GRE over IPsec combine les deux.

2. Environnement et topologie

> **Attention :** Le c7200 conseillé est remplacé par un Cisco 3725 (IOS 12.4(15)T14 Advanced Enterprise, commandes `crypto` disponibles), 256 Mo de RAM, carte WIC-2T : les interfaces s0/0 et s0/1 du schéma existent telles quelles (Serial0/0, Serial0/1).

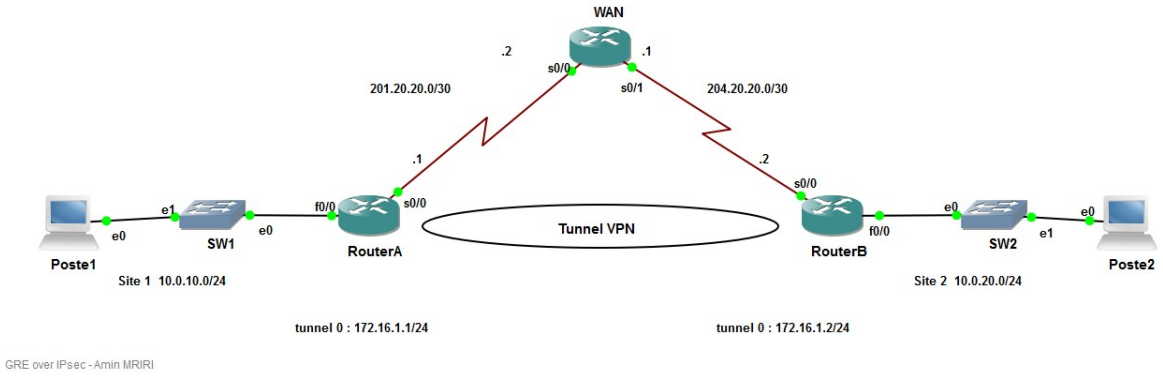


Figure 1 : Topologie GRE over IPsec sous GNS3

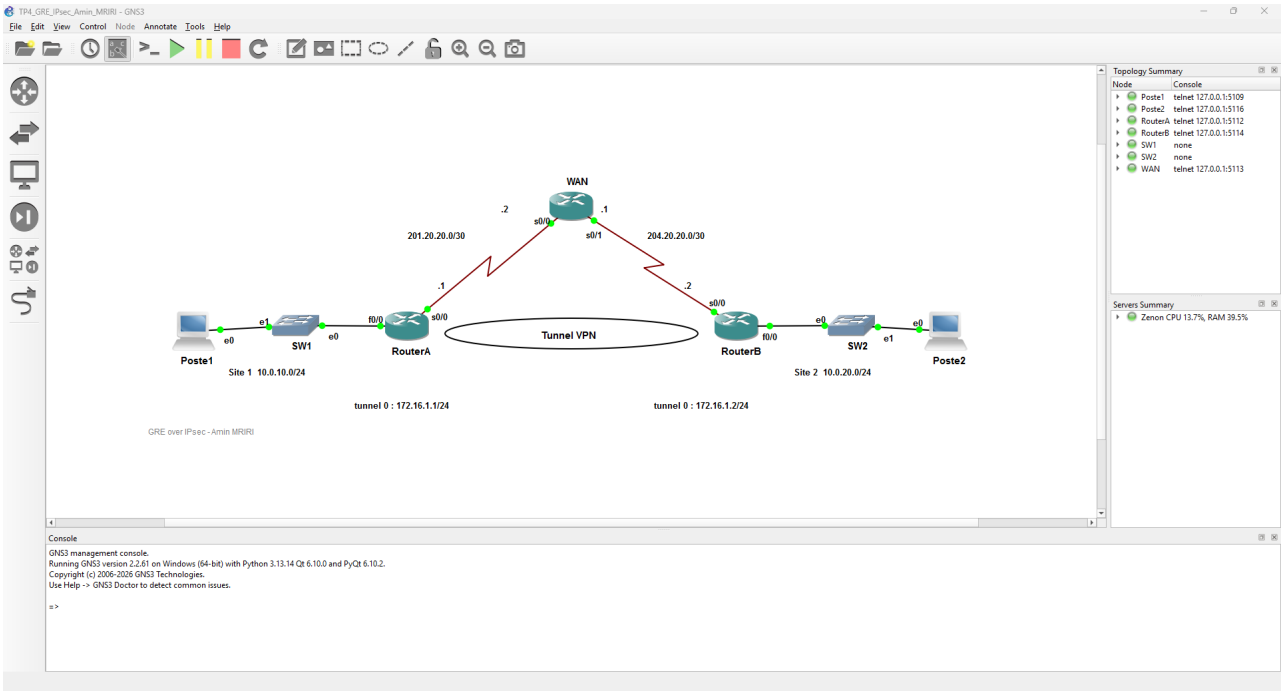


Figure 2 : Fenêtre GNS3, équipements démarrés

Équipement	Interface	Adresse
RouterA	Fa0/0 / Serial0/0	10.0.10.1/24 / 201.20.20.1/30
WAN	Serial0/0 / Serial0/1	201.20.20.2/30 / 204.20.20.1/30
RouterB	Fa0/0 / Serial0/0	10.0.20.1/24 / 204.20.20.2/30

Équipement	Interface	Adresse
RouterA / RouterB	Tunnel0	172.16.1.1/24 / 172.16.1.2/24
Poste1 / Poste2	NIC	10.0.10.10/24 / 10.0.20.10/24

3. Partie I : configuration du tunnel GRE

3.1 Configuration de base

Sur RouterA et RouterB : adresses, NAT (interfaces inside/outside, surcharge sur l'interface série) et route par défaut vers le WAN. L'ACL 100 **exclut** de la NAT le trafic d'un site vers l'autre (deny), qui doit passer par le tunnel sans être traduit.

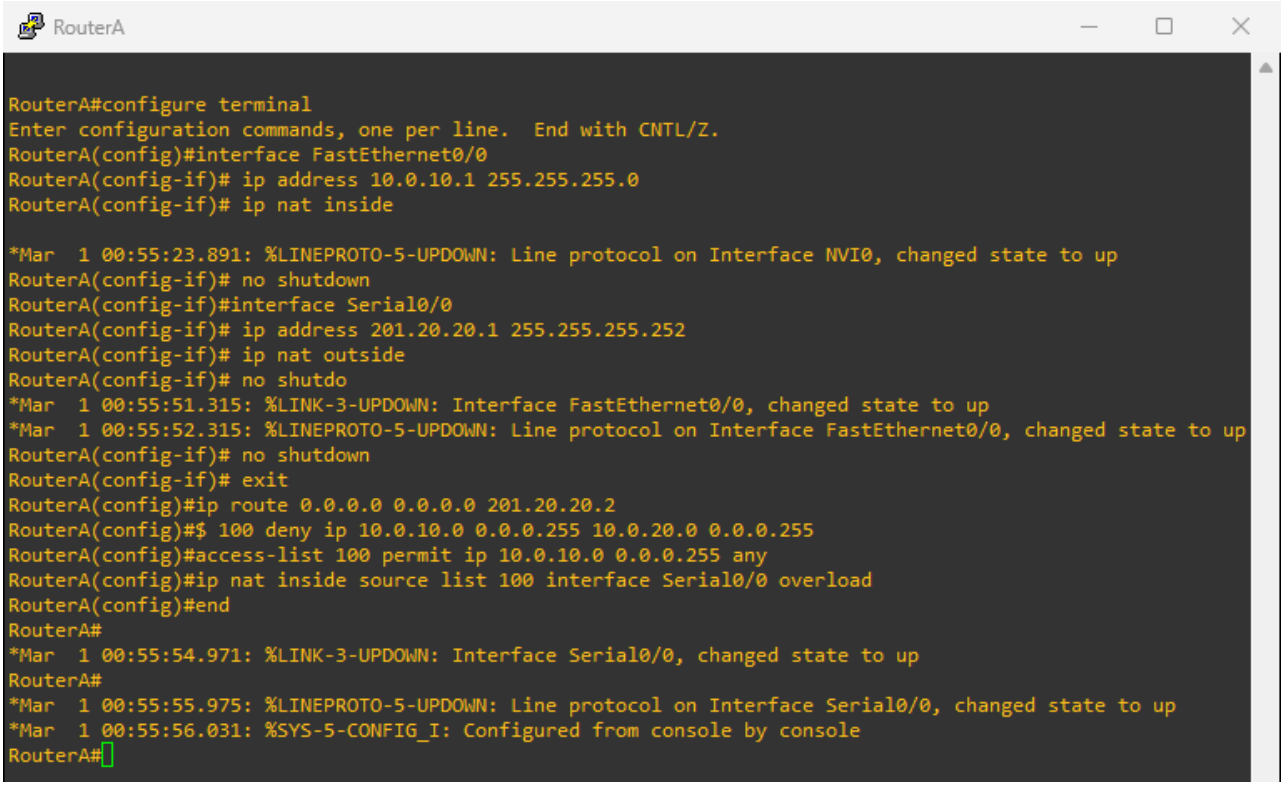


Figure 3 : Configuration de base de RouterA

WAN

```
WAN#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
WAN(config)#interface Serial0/0
WAN(config-if)# ip address 201.20.20.2 255.255.255.252
WAN(config-if)# clock rate 64000
WAN(config-if)# no shutdown
WAN(config-if)#interface Serial0/1
WAN(config-if)# ip address 204.20.20.1 255.255.255.252
WAN(config-if)# clock rate 64000
WAN(config-if)# no shutdown
WAN(config-if)#end
WAN#
*Mar  1 00:55:59.319: %LINK-3-UPDOWN: Interface Serial0/0, changed state to up
WAN#
*Mar  1 00:55:59.895: %SYS-5-CONFIG_I: Configured from console by console
*Mar  1 00:56:00.323: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
WAN#
*Mar  1 00:56:01.251: %LINK-3-UPDOWN: Interface Serial0/1, changed state to up
WAN#
*Mar  1 00:56:02.255: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to up
WAN#
```

Figure 4 : Configuration du routeur WAN

RouterB

```
RouterB#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
RouterB(config)#interface FastEthernet0/0
RouterB(config-if)# ip address 10.0.20.1 255.255.255.0
RouterB(config-if)# ip nat inside

*Mar  1 00:56:15.447: %LINEPROTO-5-UPDOWN: Line protocol on Interface NVI0, changed state to up
RouterB(config-if)# no shutdown
RouterB(config-if)#interface Serial0/0
RouterB(config-if)# ip address 204.20.20.2 255.255.255.252
RouterB(config-if)# ip nat outside
RouterB(config-if)# no shutdown
RouterB(config-if)# exit
RouterB(config)#ip route 0.0.0.0 0.0.0.0 204.20.20.1
RouterB(config)#
*Mar  1 00:56:42.799: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:56:43.799: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
RouterB(config)#$ 100 deny ip 10.0.20.0 0.0.0.255 10.0.10.0 0.0.0.255
RouterB(config)#access-list 100 permit ip 10.0.20.0 0.0.0.255 any
RouterB(config)#ip nat inside source list 100 interface Serial0/0 overload

*Mar  1 00:56:45.319: %LINK-3-UPDOWN: Interface Serial0/0, changed state to up
RouterB(config)#end
RouterB#
*Mar  1 00:56:46.327: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
*Mar  1 00:56:46.967: %SYS-5-CONFIG_I: Configured from console by console
RouterB#
```

Figure 5 : Configuration de base de RouterB

Poste2 - PuTTY

```
Poste2> ip 10.0.20.10/24 10.0.20.1
Checking for duplicate address...
PC1 : 10.0.20.10 255.255.255.0 gateway 10.0.20.1

Poste2> save
Saving startup configuration to startup.vpc
.  done

Poste2>
```

Figure 6 : Adressage de Poste2

3.2 Tunnel GRE

```
interface Tunnel 0
 ip address 172.16.1.1 255.255.255.0
 ip mtu 1400
 tunnel source S0/0
 tunnel destination 204.20.20.2
```

`ip mtu 1400` réserve de la place pour les en-têtes ajoutés par GRE (24 octets) puis par IPsec, afin d'éviter la fragmentation des paquets.

RouterA

```
RouterA#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
RouterA(config)#interface Tunnel 0
RouterA(config-if)# ip address 172.16.1.1 255.255.255.0
RouterA(config-if)# ip mtu 1400
RouterA(config-if)# tunnel source Serial0/0
RouterA(config-if)# tunnel destination 204.20.20.2
RouterA(config-if)#end
RouterA#
*Mar  1 00:57:01.887: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to down
*Mar  1 00:57:02.623: %SYS-5-CONFIG_I: Configured from console by console
RouterA#
*Mar  1 00:57:03.239: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up
RouterA#
```

Figure 7 : RouterA : interface Tunnel 0

RouterB

```
RouterB#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
RouterB(config)#interface Tunnel 0
RouterB(config-if)# ip address 172.16.1.2 255.255.255.0
RouterB(config-if)# ip mtu 1400
RouterB(config-if)# tunnel source Serial0/0
RouterB(config-if)# tunnel destination 201.20.20.1
RouterB(config-if)#end
RouterB#
*Mar  1 00:57:19.923: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to down
*Mar  1 00:57:20.663: %SYS-5-CONFIG_I: Configured from console by console
RouterB#
*Mar  1 00:57:21.319: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up
RouterB#
```

Figure 8 : RouterB : interface Tunnel 0 (destination 201.20.20.1)

RouterA

```
RouterA#$l0 | include Tunnel0|Internet|source|protocol/transport|MTU
Tunnel0 is up, line protocol is up
  Internet address is 172.16.1.1/24
  MTU 1514 bytes, BW 9 Kbit/sec, DLY 500000 usec,
  Tunnel source 201.20.20.1 (Serial0/0), destination 204.20.20.2
  Tunnel protocol/transport GRE/IP
RouterA#
```

Figure 9 : RouterA : Tunnel0 up, encapsulation GRE/IP, source et destination du tunnel

3.3 Tests de connectivité

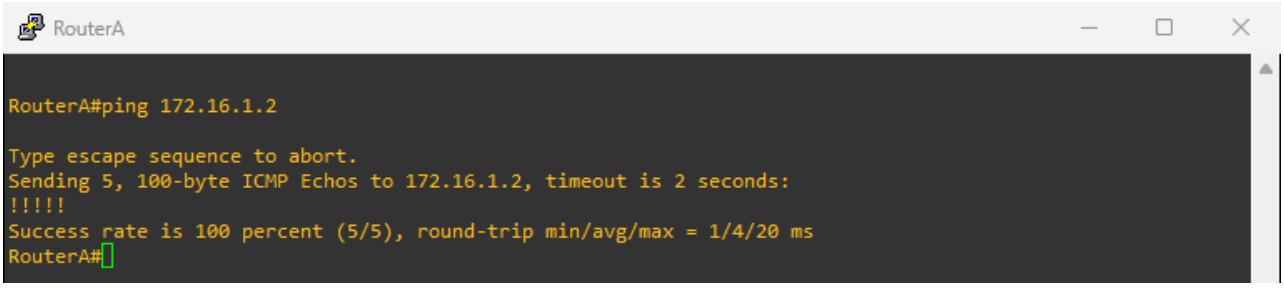


Figure 10 : RouterA#ping 172.16.1.2 : succès

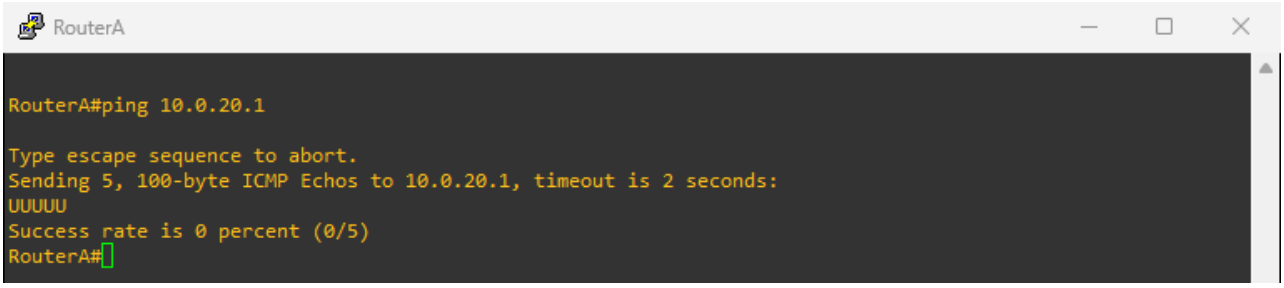


Figure 11 : RouterA#ping 10.0.20.1 : échec

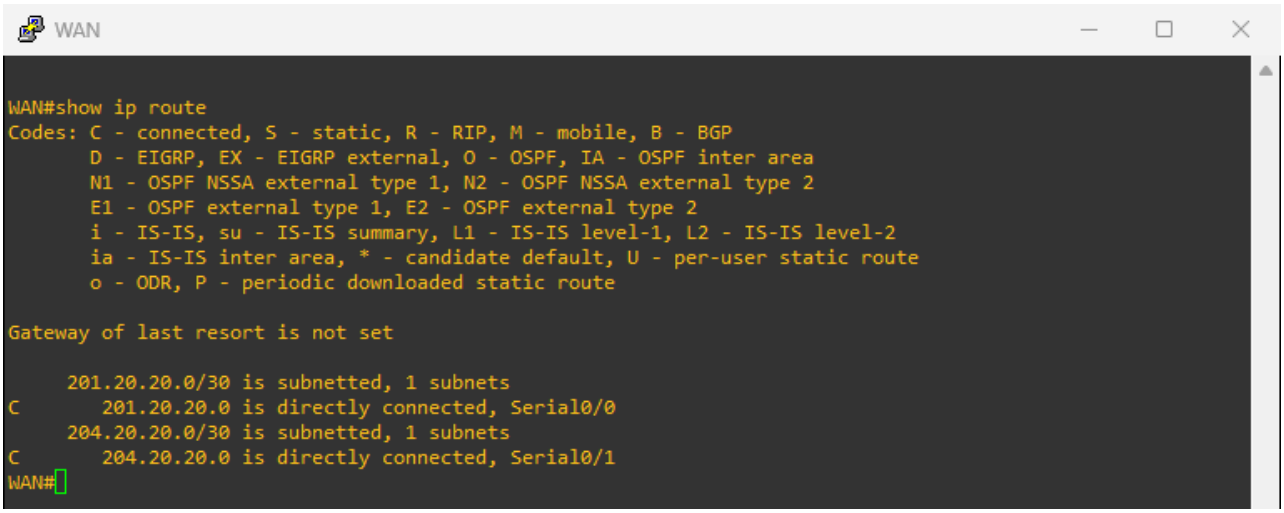


Figure 12 : WAN : seuls les réseaux publics sont connus

Question : Sur RouterA, le ping vers 10.0.20.1 aboutit-il ? Si non, pourquoi ?

> **Réponse : Non.** RouterA n'a pas de route vers 10.0.20.0/24 par le tunnel : il utilise sa route par défaut et envoie le paquet au routeur WAN.
>
> Le WAN (« Internet ») ne connaît pas les réseaux privés des sites : il renvoie un message ICMP « destination injoignable » (les « U » du ping). Il faut indiquer que le site distant se trouve derrière le tunnel.

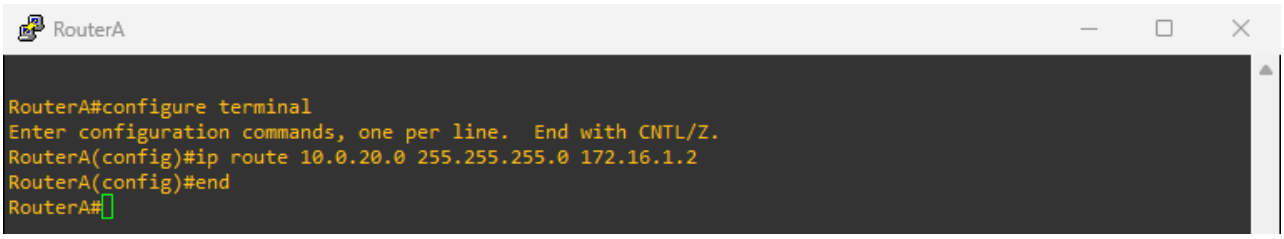


Figure 13 : RouterA : ip route 10.0.20.0 255.255.255.0 172.16.1.2

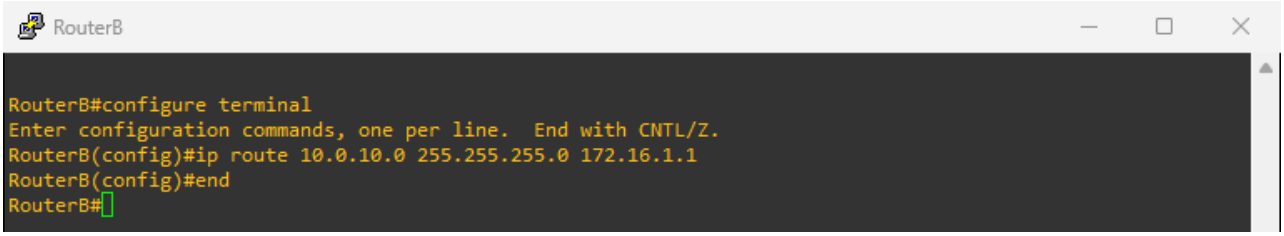


Figure 14 : RouterB : ip route 10.0.10.0 255.255.255.0 172.16.1.1

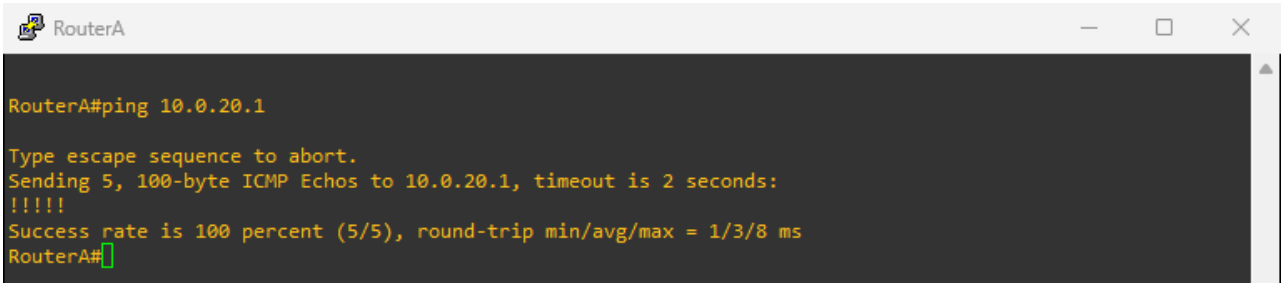


Figure 15 : RouterA#ping 10.0.20.1 : succès par le tunnel

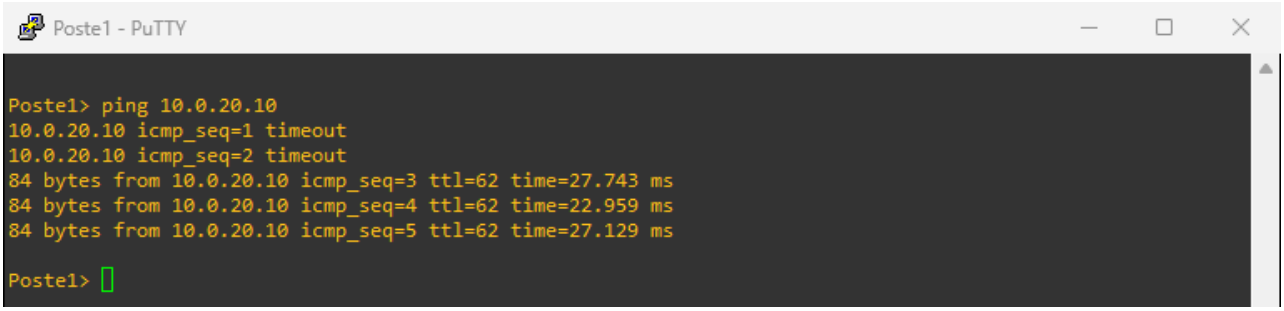


Figure 16 : Poste1 joint Poste2 à travers le tunnel GRE

3.4 Capture sur S0/0 de RouterA

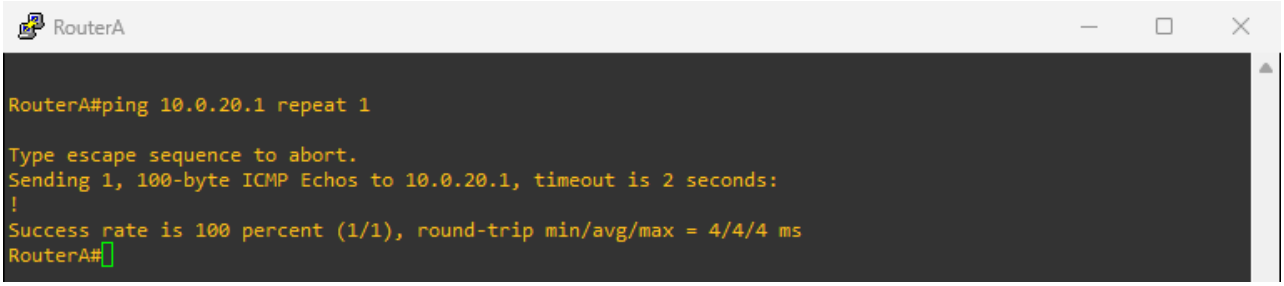


Figure 17 : RouterA#ping 10.0.20.1 repeat 1 pendant la capture

```
Frame 1: Packet, 128 bytes on wire (1024 bits), 128 bytes captured (1024 bits)
    Protocol: IP (0x0800)
Internet Protocol Version 4, Src: 201.20.20.1, Dst: 204.20.20.2
    Protocol: Generic Routing Encapsulation (47)
    Source Address: 201.20.20.1
    Destination Address: 204.20.20.2
Generic Routing Encapsulation (IP)
    Flags and Version: 0x0000
    Protocol Type: IP (0x0800)
Internet Protocol Version 4, Src: 172.16.1.1, Dst: 10.0.20.1
    Protocol: ICMP (1)
    Source Address: 172.16.1.1
    Destination Address: 10.0.20.1
Internet Control Message Protocol
```

Question : Décrire les échanges ICMP et l'encapsulation des en-têtes IP

> **Réponse :** Un seul aller-retour : une requête ICMP Echo (type 8) de 172.16.1.1 (adresse du tunnel, source choisie par RouterA) vers 10.0.20.1, puis la réponse Echo Reply (type 0).

> Sur la liaison série, chaque paquet porte **deux en-têtes IP** : l'en-tête **externe** (201.20.20.1 → 204.20.20.2, protocole 47 = GRE) qui permet de traverser le WAN, l'en-tête **GRE** (type de protocole transporté 0x0800 = IPv4), puis l'en-tête IP **interne** (172.16.1.1 → 10.0.20.1, protocole ICMP) et le message ICMP.

```
+----------------------+-----------+-----------------------+------+
| IP externe           | GRE       | IP interne            | ICMP |
| 201.20.20.1 ->       | proto     | 172.16.1.1 ->         | Echo |
| 204.20.20.2 (prot. 47)| 0x0800   | 10.0.20.1 (prot. 1)   |      |
+----------------------+-----------+-----------------------+------+
```

4. Partie II : cryptage du tunnel GRE par IPsec

IPsec ne transporte pas le multicast : on chiffre donc les paquets GRE (qui, eux, peuvent transporter un protocole de routage). L'ACL 101 sélectionne le trafic GRE entre les deux extrémités publiques du tunnel.

Élément	Configuration	Rôle
Politique IKE	`crypto isakmp policy 1` / `authentication pre-share`	phase 1 : authentification par clé partagée
Clé	`crypto isakmp key 6 P@ssword address <pair>`	clé P@ssword associée à l'adresse du pair
Durée de vie	`crypto ipsec security-association lifetime seconds 4096`	renouvellement des clés IPsec
Transform-set	`strong` : esp-3des esp-md5-hmac	chiffrement 3DES, intégrité MD5
ACL 101	`permit gre host 201.20.20.1 host 204.20.20.2`	trafic GRE à chiffrer
Crypto map	`S2SVPN 10 ipsec-isakmp` sur l'interface série	lie ACL, pair et transform-set

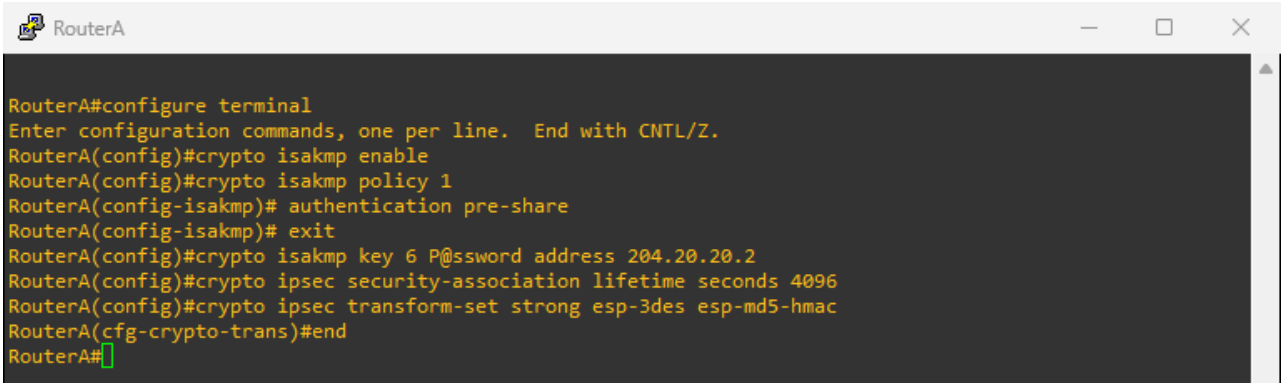


Figure 18 : RouterA : ISAKMP, politique, clé et transform-set

RouterA

```
RouterA#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
RouterA(config)#
*Mar  1 00:57:56.043: %SYS-5-CONFIG_I: Configured from console by console
RouterA(config)#access-list 101 permit gre host 201.20.20.1 host 204.20.20.2
RouterA(config)#crypto map S2SVPN 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
RouterA(config-crypto-map)# match address 101
RouterA(config-crypto-map)# set peer 204.20.20.2
RouterA(config-crypto-map)# set transform-set strong
RouterA(config-crypto-map)# exit
RouterA(config)#interface Serial0/0
RouterA(config-if)# crypto map S2SVPN
RouterA(config-if)#end
RouterA#
*Mar  1 00:58:00.707: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
RouterA#
*Mar  1 00:58:01.023: %SYS-5-CONFIG_I: Configured from console by console
RouterA#
```

Figure 19 : RouterA : ACL 101, crypto map S2SVPN et application sur l'interface S0/0

RouterB

```
RouterB#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
RouterB(config)#crypto isakmp enable
RouterB(config)#crypto isakmp policy 1
RouterB(config-isakmp)# authentication pre-share
RouterB(config-isakmp)# exit
RouterB(config)#crypto isakmp key 6 P@ssword address 201.20.20.1
RouterB(config)#crypto ipsec security-association lifetime seconds 4096
RouterB(config)#crypto ipsec transform-set strong esp-3des esp-md5-hmac
RouterB(cfg-crypto-trans)#end
RouterB#
```

Figure 20 : RouterB : même configuration, pair 201.20.20.1

RouterB

```
RouterB#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
RouterB(config)#
*Mar  1 00:58:17.783: %SYS-5-CONFIG_I: Configured from console by console
RouterB(config)#access-list 101 permit gre host 204.20.20.2 host 201.20.20.1
RouterB(config)#crypto map S2SVPN 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
RouterB(config-crypto-map)# match address 101
RouterB(config-crypto-map)# set peer 201.20.20.1
RouterB(config-crypto-map)# set transform-set strong
RouterB(config-crypto-map)# exit
RouterB(config)#interface Serial0/0
RouterB(config-if)# crypto map S2SVPN
RouterB(config-if)#end
RouterB#
*Mar  1 00:58:22.575: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
RouterB#
*Mar  1 00:58:22.907: %SYS-5-CONFIG_I: Configured from console by console
RouterB#
```

Figure 21 : RouterB : ACL 101 inversée et crypto map

4.1 Tests via le tunnel GRE over IPsec

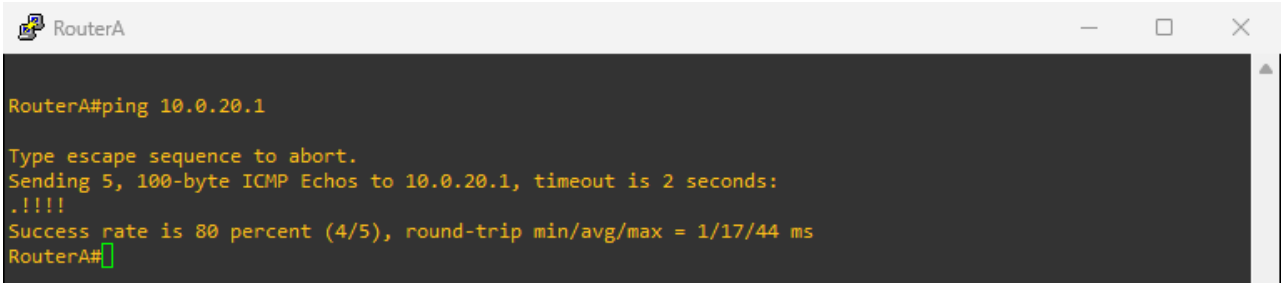


Figure 22 : RouterA#ping 10.0.20.1 : succès, le trafic GRE est maintenant chiffré

Capture sur S0/0 de RouterA : négociation ISAKMP puis uniquement des paquets ESP ; le GRE n'est plus visible en clair.

```
1   0.000000  201.20.20.1 → 204.20.20.2  ISAKMP 196 Identity Protection (Main Mode)
2   0.033329  204.20.20.2 → 201.20.20.1  ISAKMP 136 Identity Protection (Main Mode)
3   0.050007  201.20.20.1 → 204.20.20.2  ISAKMP 304 Identity Protection (Main Mode)
4   0.066690  204.20.20.2 → 201.20.20.1  ISAKMP 304 Identity Protection (Main Mode)
5   0.077749  201.20.20.1 → 204.20.20.2  ISAKMP 132 Identity Protection (Main Mode)
6   0.088871  204.20.20.2 → 201.20.20.1  ISAKMP 100 Identity Protection (Main Mode)
7   0.088871  201.20.20.1 → 204.20.20.2  ISAKMP 196 Quick Mode
8   0.099952  204.20.20.2 → 201.20.20.1  ISAKMP 196 Quick Mode
9   0.105543  201.20.20.1 → 204.20.20.2  ISAKMP 92 Quick Mode
10   2.018420  201.20.20.1 → 204.20.20.2  ESP 180 ESP (SPI=0xd5fe523b)
11   2.022232  204.20.20.2 → 201.20.20.1  ESP 180 ESP (SPI=0x2c011663)
12   2.027796  201.20.20.1 → 204.20.20.2  ESP 180 ESP (SPI=0xd5fe523b)
13   2.027796  204.20.20.2 → 201.20.20.1  ESP 180 ESP (SPI=0x2c011663)
14   2.027796  201.20.20.1 → 204.20.20.2  ESP 180 ESP (SPI=0xd5fe523b)

Frame 10: Packet, 180 bytes on wire (1440 bits), 180 bytes captured (1440 bits)
    Protocol: IP (0x0800)
Internet Protocol Version 4, Src: 201.20.20.1, Dst: 204.20.20.2
    Protocol: Encap Security Payload (50)
    Source Address: 201.20.20.1
    Destination Address: 204.20.20.2
Encapsulating Security Payload
    ESP SPI: 0xd5fe523b (3590214203)
    ESP Sequence: 1
```

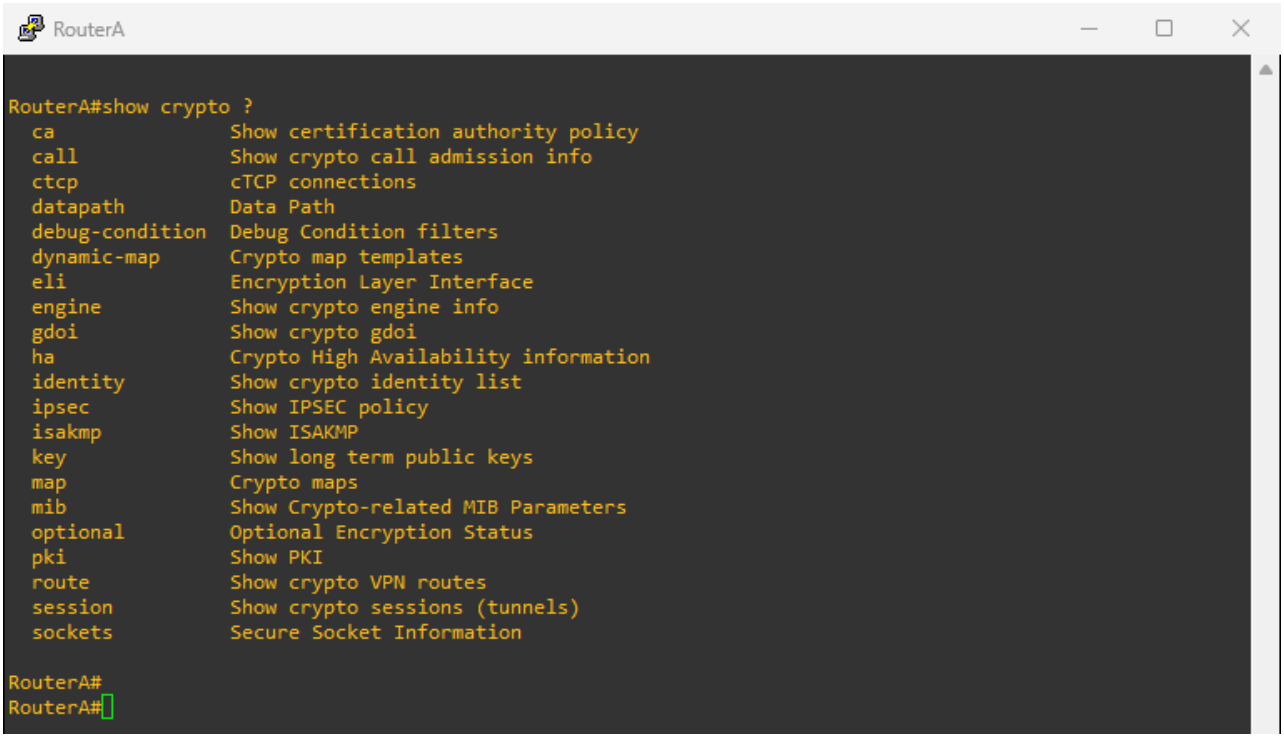


Figure 23 : RouterA#show crypto ? : liste des informations consultables

RouterA

```
RouterA#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
204.20.20.2     201.20.20.1     QM_IDLE           1001    0 ACTIVE

IPv6 Crypto ISAKMP SA

RouterA#
```

Figure 24 : show crypto isakmp sa : SA IKE avec 204.20.20.2

RouterA

```
RouterA#show crypto ipsec sa

interface: Serial0/0
    Crypto map tag: S2SVPN, local addr 201.20.20.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (201.20.20.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (204.20.20.2/255.255.255.255/47/0)
   current_peer 204.20.20.2 port 500
     PERMIT, flags={origin_is_acl,ipsec_sa_request_sent}
    #pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 7
    #pkts decaps: 7, #pkts decrypt: 7, #pkts verify: 7
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 1, #recv errors 0

     local crypto endpt.: 201.20.20.1, remote crypto endpt.: 204.20.20.2
     path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0
     current outbound spi: 0xD5FE523B(3590214203)

     inbound esp sas:
      spi: 0x2C011663(738268771)
        transform: esp-3des esp-md5-hmac ,
        in use settings ={Tunnel, }
        conn id: 1, flow_id: SW:1, crypto map: S2SVPN
        sa timing: remaining key lifetime (k/sec): (4463467/4084)
        IV size: 8 bytes
        replay detection support: Y
        Status: ACTIVE

     inbound ah sas:

     inbound pcp sas:

     outbound esp sas:
      spi: 0xD5FE523B(3590214203)
        transform: esp-3des esp-md5-hmac ,
        in use settings ={Tunnel, }
        conn id: 2, flow_id: SW:2, crypto map: S2SVPN
        sa timing: remaining key lifetime (k/sec): (4463467/4084)
        IV size: 8 bytes
        replay detection support: Y
        Status: ACTIVE

     outbound ah sas:

     outbound pcp sas:
RouterA#
```

Figure 25 : show crypto ipsec sa : SA IPsec et compteurs de paquets chiffrés

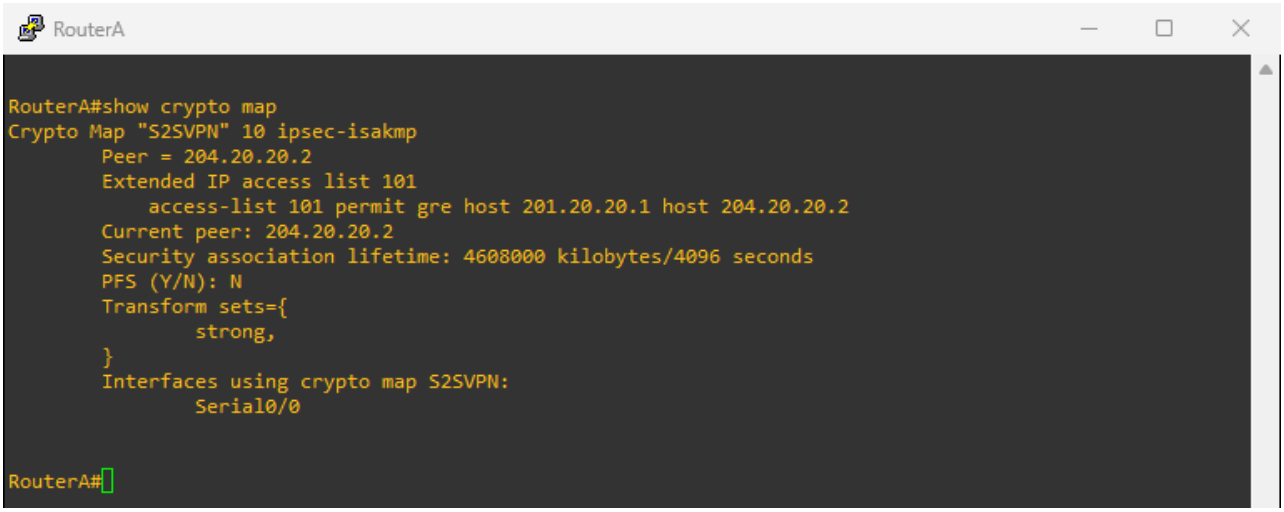


Figure 26 : show crypto map : crypto map S2SVPN, pair, ACL 101 et transform-set

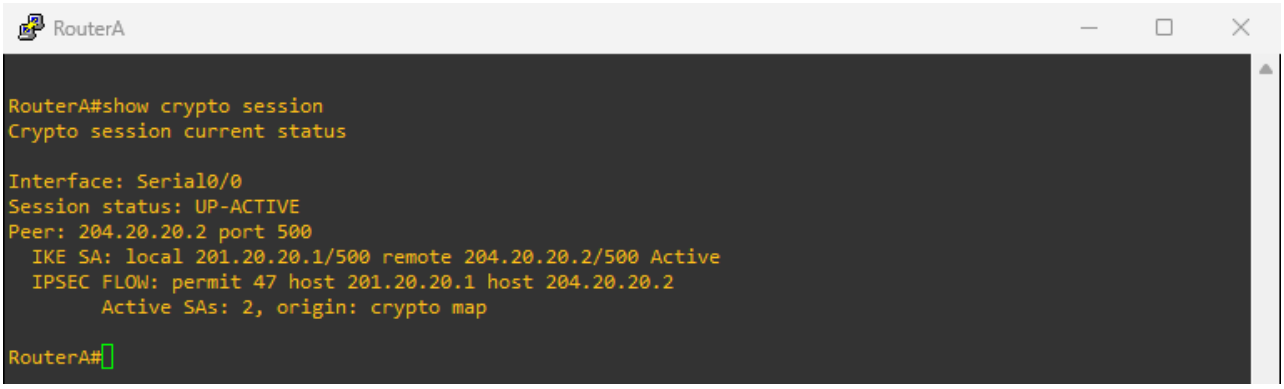


Figure 27 : show crypto session : session IKE et IPsec active

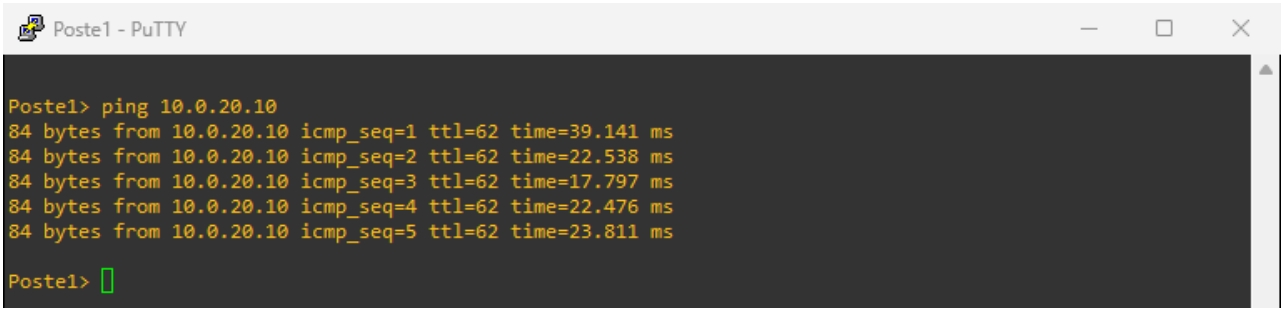


Figure 28 : Poste1 joint Poste2 : trafic chiffré de bout en bout entre les sites

Encapsulation finale sur le WAN : nouvel en-tête IP (201.20.20.1 → 204.20.20.2, protocole 50 ESP) + en-tête ESP + [paquet GRE complet chiffré : IP externe GRE + GRE + IP interne + données] + fin ESP / authentification. Un observateur sur le WAN ne voit plus que les adresses publiques des deux routeurs.

5. Conclusion

Le tunnel GRE relie les deux sites comme un lien point à point (réseau 172.16.1.0/24) à travers le WAN, qui ignore les réseaux privés. Les routes statiques vers le tunnel orientent le trafic inter-sites, et l'ACL de NAT empêche sa traduction. IPsec chiffre ensuite les paquets GRE (ESP 3DES/MD5) : on obtient un VPN capable de transporter du multicast et un protocole de routage, tout en restant confidentiel.

6. Annexe : configurations

Routeur RouterA

```
version 12.4
hostname RouterA
crypto isakmp policy 1
 authentication pre-share
!
crypto isakmp key 6 P@ssword address 204.20.20.2
!
crypto ipsec security-association lifetime seconds 4096
!
crypto ipsec transform-set strong esp-3des esp-md5-hmac
!
crypto map S2SVPN 10 ipsec-isakmp
 set peer 204.20.20.2
 set transform-set strong
 match address 101
!
interface Tunnel0
 ip address 172.16.1.1 255.255.255.0
 ip mtu 1400
 tunnel source Serial0/0
 tunnel destination 204.20.20.2
!
interface FastEthernet0/0
 ip address 10.0.10.1 255.255.255.0
 ip nat inside
 ip virtual-reassembly
!
interface Serial0/0
 ip address 201.20.20.1 255.255.255.252
 ip nat outside
 ip virtual-reassembly
 clock rate 2000000
 crypto map S2SVPN
!
ip route 0.0.0.0 0.0.0.0 201.20.20.2
ip route 10.0.20.0 255.255.255.0 172.16.1.2
ip nat inside source list 100 interface Serial0/0 overload
access-list 100 deny   ip 10.0.10.0 0.0.0.255 10.0.20.0 0.0.0.255
access-list 100 permit ip 10.0.10.0 0.0.0.255 any
access-list 101 permit gre host 201.20.20.1 host 204.20.20.2
crypto isakmp policy 1
 authentication pre-share
crypto isakmp key 6 P@ssword address 204.20.20.2
crypto ipsec security-association lifetime seconds 4096
crypto ipsec transform-set strong esp-3des esp-md5-hmac
crypto map S2SVPN 10 ipsec-isakmp
 set peer 204.20.20.2
 set transform-set strong
 match address 101
ip route 0.0.0.0 0.0.0.0 201.20.20.2
ip route 10.0.20.0 255.255.255.0 172.16.1.2
ip nat inside source list 100 interface Serial0/0 overload
access-list 100 deny   ip 10.0.10.0 0.0.0.255 10.0.20.0 0.0.0.255
access-list 100 permit ip 10.0.10.0 0.0.0.255 any
access-list 101 permit gre host 201.20.20.1 host 204.20.20.2
```

Routeur WAN

```
version 12.4
hostname WAN
interface Serial0/0
 ip address 201.20.20.2 255.255.255.252
 clock rate 64000
!
interface Serial0/1
 ip address 204.20.20.1 255.255.255.252
 clock rate 64000
!
```

Routeur RouterB

```
version 12.4
hostname RouterB
crypto isakmp policy 1
 authentication pre-share
!
crypto isakmp key 6 P@ssword address 201.20.20.1
!
crypto ipsec security-association lifetime seconds 4096
!
crypto ipsec transform-set strong esp-3des esp-md5-hmac
!
crypto map S2SVPN 10 ipsec-isakmp
 set peer 201.20.20.1
 set transform-set strong
 match address 101
!
interface Tunnel0
 ip address 172.16.1.2 255.255.255.0
 ip mtu 1400
 tunnel source Serial0/0
 tunnel destination 201.20.20.1
!
interface FastEthernet0/0
 ip address 10.0.20.1 255.255.255.0
 ip nat inside
 ip virtual-reassembly
!
interface Serial0/0
 ip address 204.20.20.2 255.255.255.252
 ip nat outside
 ip virtual-reassembly
 clock rate 2000000
 crypto map S2SVPN
!
ip route 0.0.0.0 0.0.0.0 204.20.20.1
ip route 10.0.10.0 255.255.255.0 172.16.1.1
ip nat inside source list 100 interface Serial0/0 overload
access-list 100 deny   ip 10.0.20.0 0.0.0.255 10.0.10.0 0.0.0.255
access-list 100 permit ip 10.0.20.0 0.0.0.255 any
access-list 101 permit gre host 204.20.20.2 host 201.20.20.1
crypto isakmp policy 1
 authentication pre-share
crypto isakmp key 6 P@ssword address 201.20.20.1
crypto ipsec security-association lifetime seconds 4096
crypto ipsec transform-set strong esp-3des esp-md5-hmac
crypto map S2SVPN 10 ipsec-isakmp
 set peer 201.20.20.1
 set transform-set strong
 match address 101
ip route 0.0.0.0 0.0.0.0 204.20.20.1
ip route 10.0.10.0 255.255.255.0 172.16.1.1
ip nat inside source list 100 interface Serial0/0 overload
access-list 100 deny   ip 10.0.20.0 0.0.0.255 10.0.10.0 0.0.0.255
access-list 100 permit ip 10.0.20.0 0.0.0.255 any
access-list 101 permit gre host 204.20.20.2 host 201.20.20.1
```